

SP_EDA

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```
data2 <- read_csv("SP_Data.csv")
```

```
## Rows: 34945 Columns: 29
## -- Column specification -----
## Delimiter: ","
## chr (24): Incident PSAP Call Date Time (eTimes.01), Incident Unit En Route D...
## dbl (3): Incident City GNIS (eScene.17), Scene GPS Latitude (eScene.11), Sc...
## lgl (2): Scene US National Grid Coordinates (eScene.12), Date/Time of Sympt...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
#Look at top reasons
```

```
data2|>
  group_by(`Dispatch Reason (eDispatch.01)`)|>
  summarise(n = n())|>
  ungroup()|>
  arrange(desc(n))|>
  slice_head(n = 10)
```

```
## # A tibble: 10 x 2
##   `Dispatch Reason (eDispatch.01)`      n
##   <chr>                                <int>
## 1 Fall(s)                             4313
## 2 Breathing Problem                   3528
## 3 Transfer / Interfacility / Palliative Care 3221
## 4 Sick Person                         3093
## 5 MVC / Traffic / Transportation Incident 2666
## 6 Chest Pain (Non-Traumatic)           2193
## 7 Abdominal Pain / Problems            1454
## 8 Convulsions / Seizure                 964
## 9 No Other Appropriate Choice           951
## 10 Pain not otherwise specified         848
```

```
#want to group by every month of every year
```

```
dates <- data2 |>
  mutate(
    date = mdy_hms(`Incident Date Time`),
    `mm/yyyy` = format(date, "%m/%Y"),
    month = my(`mm/yyyy`),
```

```

season = case_when(
  month(date) %in% c(12, 1, 2, 3) ~ (year(date) - 2021) * 4 + 1,
  month(date) %in% c(4, 5) ~ (year(date) - 2021) * 4 + 2,
  month(date) %in% c(6, 7, 8) ~ (year(date) - 2021) * 4 + 3,
  month(date) %in% c(9, 10, 11) ~ (year(date) - 2021) * 4 + 4
),
season_label = case_when(
  month(date) %in% c(12, 1, 2, 3) ~ paste("Winter", year(date)),
  month(date) %in% c(4, 5) ~ paste("Spring", year(date)),
  month(date) %in% c(6, 7, 8) ~ paste("Summer", year(date)),
  month(date) %in% c(9, 10, 11) ~ paste("Fall", year(date))
)
)|>
  arrange(date)
#finding top reasons
top_reasons <- dates|>
  count(`Dispatch Reason (eDispatch.01)`, sort = TRUE)|>
  slice_head(n = 5)

top_12_reasons <- dates |>
  count(`Dispatch Reason (eDispatch.01)`, sort = TRUE)|>
  slice_head(n = 12)

onlyTopReasons <- dates|>
  semi_join(top_reasons, by = "Dispatch Reason (eDispatch.01)")

onlyTop12Reasons <- dates |>
  semi_join(top_12_reasons, by = "Dispatch Reason (eDispatch.01)")

#grouped by month
byMonth <- onlyTopReasons |>
  count(`Dispatch Reason (eDispatch.01)`, `mm/yyyy`) |>
  mutate(month = my(`mm/yyyy`)) |>
  ggplot(aes(
    x = month,
    y = n,
    color = `Dispatch Reason (eDispatch.01)`
  )) +
  geom_line()
#Top Reasons by season
bySeason <- onlyTopReasons |>
  count(`Dispatch Reason (eDispatch.01)`, season, season_label) |>
  ggplot(aes(
    x = season,
    y = n,
    color = `Dispatch Reason (eDispatch.01)`,
    group = `Dispatch Reason (eDispatch.01)`
  )) +
  geom_line(linewidth = 1.1) +
  scale_x_continuous(
    breaks = seq(1, max(dates$season), by = 2),
    labels = unique(dates$season_label)[seq(1, length(unique(dates$season_label)), by = 2)]
  )+

```

```

theme(
  axis.text.x = element_text(
    angle = 45,
    hjust = 1))+
labs(title = "Top Reasons Over the Seasons",
     x = "Season",
     y = "Number of Dispatches")

```

```

overallCounts <- onlyTop12Reasons |>
  count(season, season_label)
#Top Reasons by season but facet wrapped
fw_by_season <- onlyTop12Reasons |>
  count(`Dispatch Reason (eDispatch.01)`, season, season_label) |>
  ggplot(mapping = aes(
    x = season,
    y = n,
    color = `Dispatch Reason (eDispatch.01)`
  )) +
  geom_line(linewidth = 1.1) +
  geom_line(data = overallCounts, aes(x = season, y = n),
            color = "black") +
  scale_x_continuous(
    breaks = seq(1, max(onlyTop12Reasons$season), by = 4),
    labels = unique(onlyTop12Reasons$season_label)[seq(1, length(unique(onlyTop12Reasons$season_label))
  )]+
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1))+
  labs(title = "Top Reasons Over the Seasons",
       x = "Season",
       y = "Number of Dispatches") +
  facet_wrap(~`Dispatch Reason (eDispatch.01)`)

```

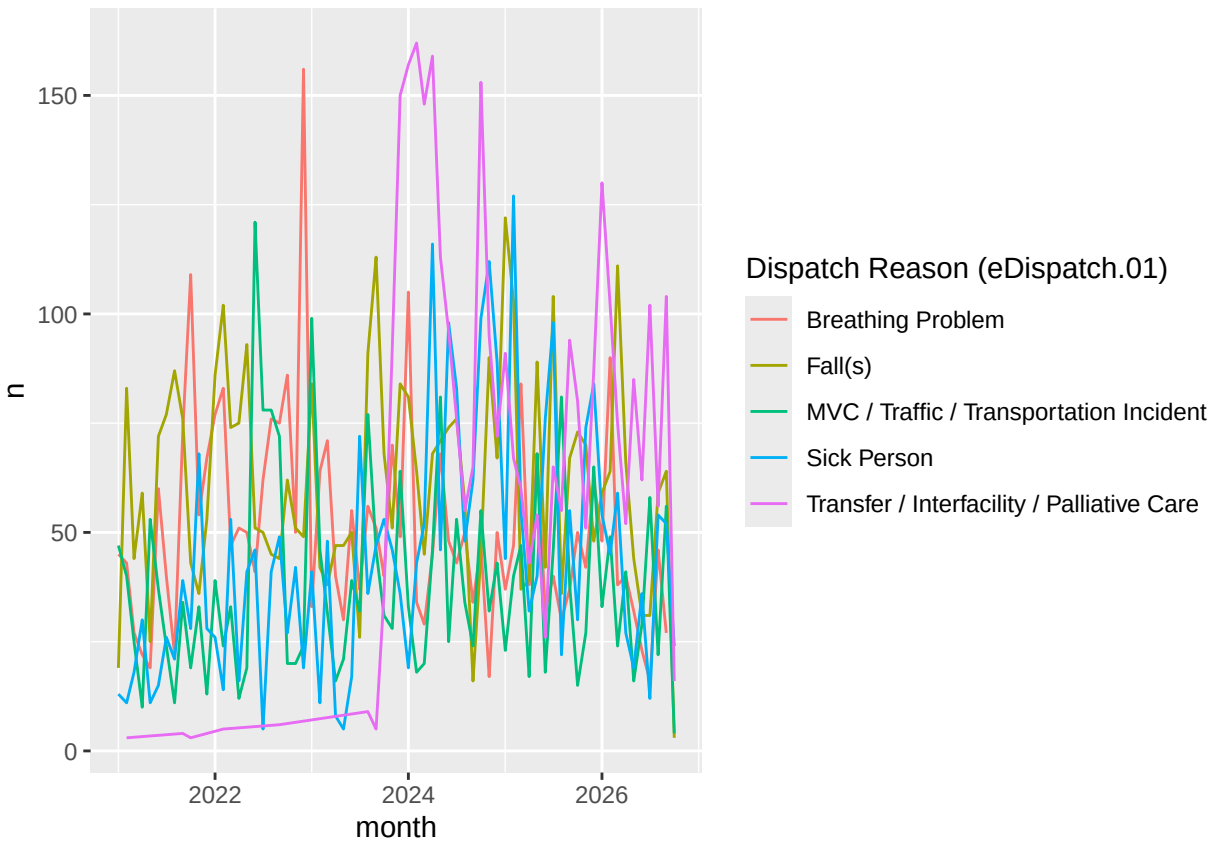
```

overallCountsMonth <- onlyTop12Reasons |>
  count(`mm/yyyy`, month)
#Top Reasons by month but facet wrapped
fw_by_month <- onlyTop12Reasons |>
  count(`Dispatch Reason (eDispatch.01)`, `mm/yyyy`, month) |>
  ggplot(mapping = aes(
    x = month,
    y = n,
    color = `Dispatch Reason (eDispatch.01)`
  )) +
  geom_line(linewidth = 1.1) +
  geom_line(data = overallCountsMonth, aes(x = month, y = n),
            color = "black") +
  scale_x_date(
    date_breaks = "6 months",
    date_labels = "%m/%Y"
  )

```

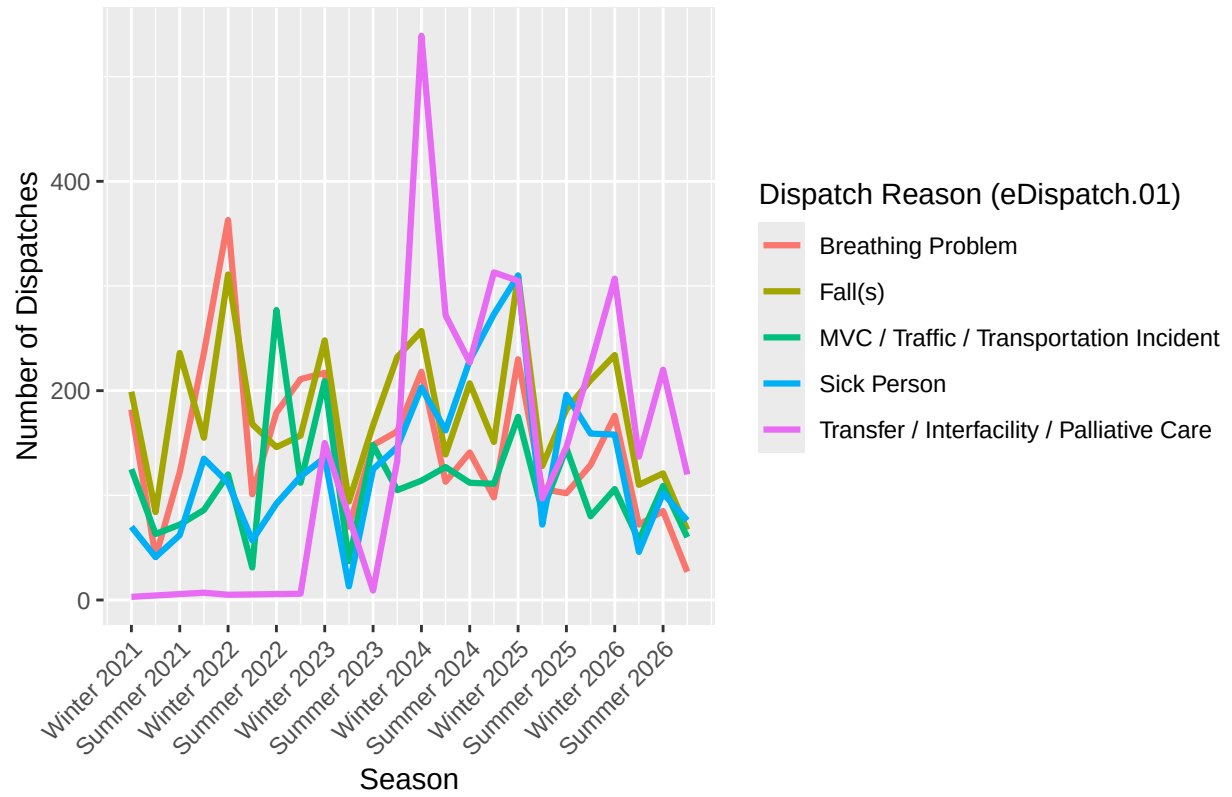
```
)+
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1))+
  labs(title = "Top Reasons Over the Months",
        x = "Month",
        y = "Number of Dispatches") +
  facet_wrap(~`Dispatch Reason (eDispatch.01)`)
```

byMonth



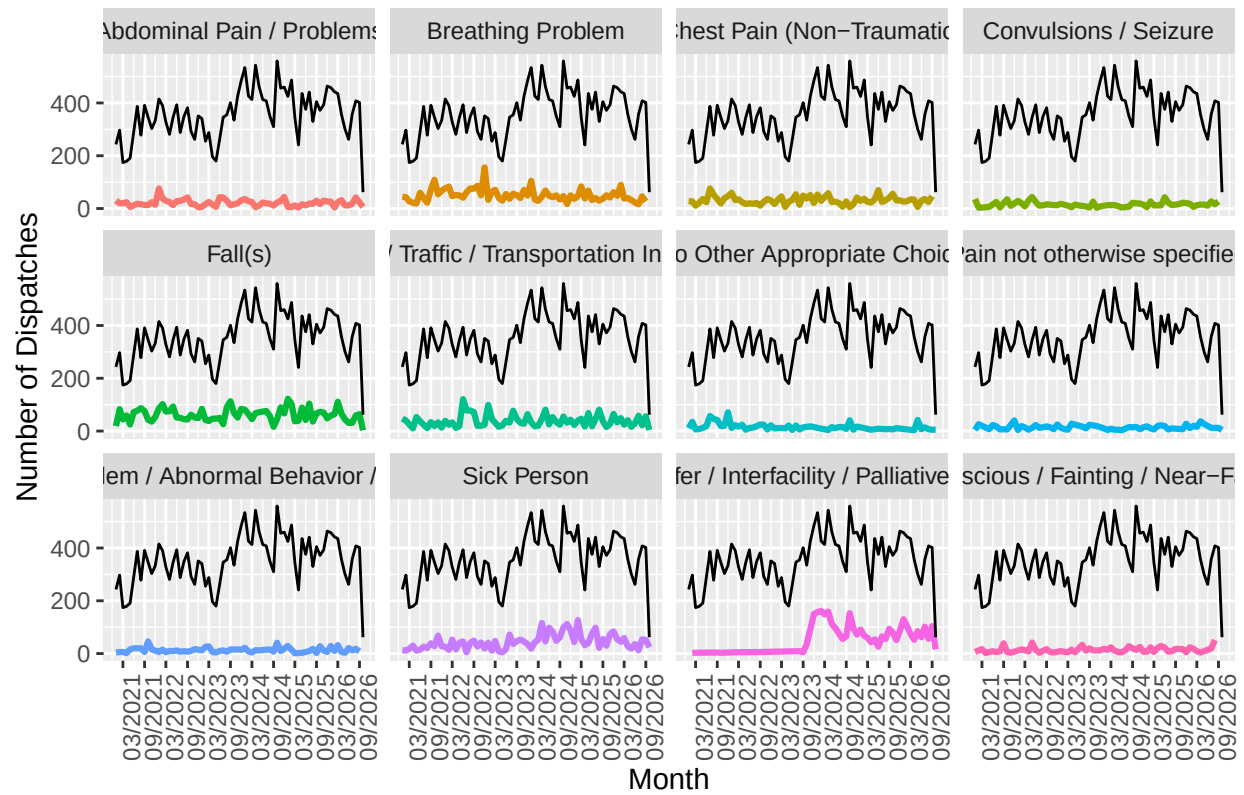
bySeason

Top Reasons Over the Seasons



fw_by_month

Top Reasons Over the Months



fw_by_season

Top Reasons Over the Seasons

